DATA STRUCTURES LAB RECORD

Array Implementation of List ADT in C

Source Code (ARRAY_LIST_ADT.C)

```
#include<stdio.h>
#define MAXSIZE 50

int arr[MAXSIZE], index = 0, nofelements;

void append(int data) {
    arr[index] = data;
    index++;
    nofelements++;
}

void traverse() {
    for (index = 0; index < nofelements; index++) {
        printf("%d->", arr[index]);
    }
}

void prepend(int data) {
    int i;
    for (i = index; i > 0; i--) {
        arr[i] = arr[i - 1];
    }
    arr[0] = data;
    index++;
    nofelements++;
}

void insert(int data, int pos) {
    int i;
    for (i = index; i > pos; i--) {
        arr[i] = arr[i - 1];
    }
    arr[pos] = data;
    index++;
    nofelements++;
}

int getnofelements() {
    return nofelements;
}

int main() {
    int n;
    append(5);
    append(6);
    append(7);
    prepend(8);
    prepend(9);
    insert(4, 2);
    traverse();
    n = getnofelements();
    printf("nof elements:%d", n);
    return 0;
}
```

Sample Output

```
9->8->4->5->6->7->nof elements:6
```